

ZACH MOHR

Mechanical Engineer | Masters of Innovation Design (May 2026)

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I bridge mechanical engineering with design thinking, currently working in aircraft interiors R&D at Textron Aviation. My background spans manufacturing optimization, product development, and hands-on fabrication. I am completing a Masters of Innovation Design at Wichita State University (May 2026) and hold a BS in Mechanical Engineering from South Dakota School of Mines & Technology. My work spans CAD, rapid prototyping, lean manufacturing, and forming simulation — from welding aluminum stages to launching patent-pending consumer products.

CORE CAPABILITIES

Mechanical Design: Linkage design, structural analysis, DFM, FMEA, forming simulation (Altair HyperForm)

Manufacturing: Process engineering, lean line balancing, time study, standard work, fixture design

CAD: CATIA V6/V5, PTC Creo, Fusion 360, SolidWorks, Inventor

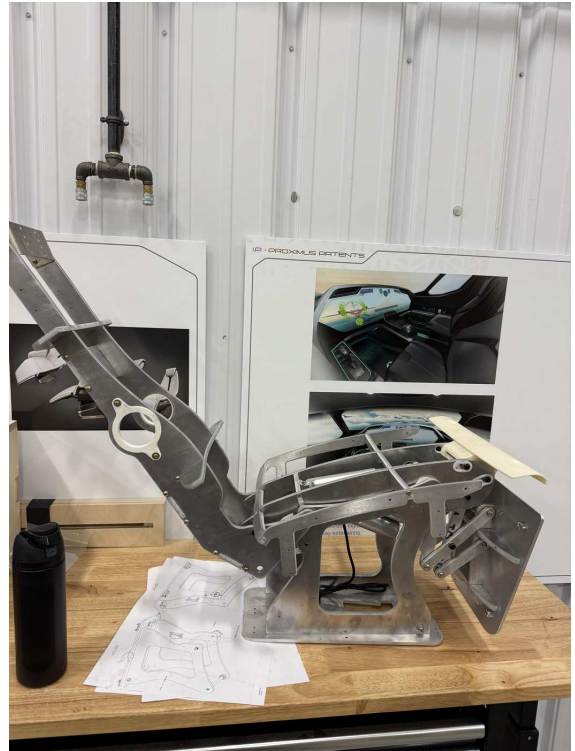
Prototyping & Fabrication: Sheet metal, CNC machining, 3D printing, silicone casting, MIG/TIG welding, brazing, lathe operation

Robotics & Electronics: Sensor integration, motor controllers, embedded systems, computer vision

PROJECT 01

Zero-G Electric Aircraft Seat

Aircraft Interiors · Textron Aviation · 2024–2025

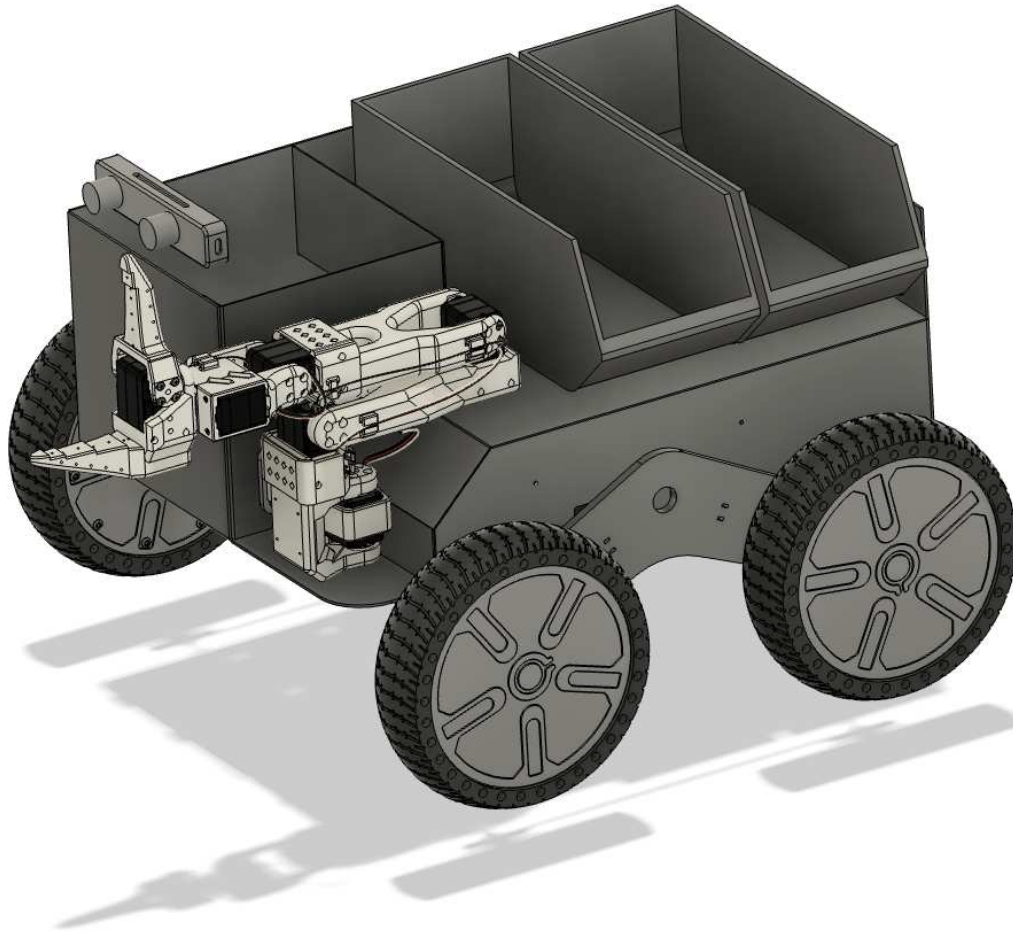


Designed a fully electric zero-gravity concept seat for a business jet interior at Textron Aviation. The seat uses three linear actuators: one for recline, one for the thighrest bolster, and one for the footrest. Designed the complete seat system including structural pans, exterior shrouding, and the underlying sheet metal structure. Engineered a custom 6-bar linkage from scratch for the footrest deployment, with the seat pan dropping in coordination with the recline mechanism. Also designed the electronics system powering the seat. This project generated two patent filings through Textron Aviation.

Sheet Metal Design, Linear Actuators, 6-Bar Linkage, Electronics, Patent (x2)

Span Robotics — Autonomous Trash Rover

Master's Thesis · Robotics · 2025–2026 (In Progress)



Master's project building an autonomous rover designed to pick up trash in outdoor spaces. The project went through several concept iterations before arriving at the current architecture, funded by an awarded innovation grant. The rover uses ODrive Robotics motor controller boards and brushless motors for the 4WD drivetrain, a ZED Mini stereo camera for depth perception and obstacle avoidance, an SO-101 robot arm for trash pickup and sorting, and a Jetson Orin Nano dev kit running the autonomy stack. The body will be fabricated from sheet metal. Build is actively in progress.

Autonomy, Computer Vision, ODrive, Jetson Orin Nano, ZED Stereo Camera, SO-101 Arm, Sheet Metal

PROJECT 03

BeethoVAN — Modular Concert Stages

Senior Capstone · April 2023



Senior capstone at SD Mines partnering with Rushmore Music Festival. Designed and built a modular aluminum stage system from scratch for a mobile concert trailer. Performed structural load analysis using composite beam theory and FEA validation. Designed height-adjustable legs with cam lock handles from laminated sheet metal. Built an articulating piano platform with linear actuator for vertical transport storage. BeethoVAN began touring summer 2023.

Mechanical Engineering, FEA, Aluminum Welding, Composite Beam Theory

Aircraft Rib Wrinkle Prediction

Aerospace Internship · June 2022



Aerospace internship project predicting and mitigating wrinkle formation in hydroformed aluminum aircraft ribs. Ran forming simulations in Altair HyperForm to predict wrinkle initiation, then designed relief cutouts into tooling to redistribute material flow. Validated the simulation approach by 3D-printing a mandrel and forming an aluminum rib, confirming the predicted wrinkle behavior matched physical results.

FEA, Forming Simulation, Altair HyperForm, Manufacturing

Bobcat Manufacturing Engineering Co-op

Manufacturing Internship · October 2021



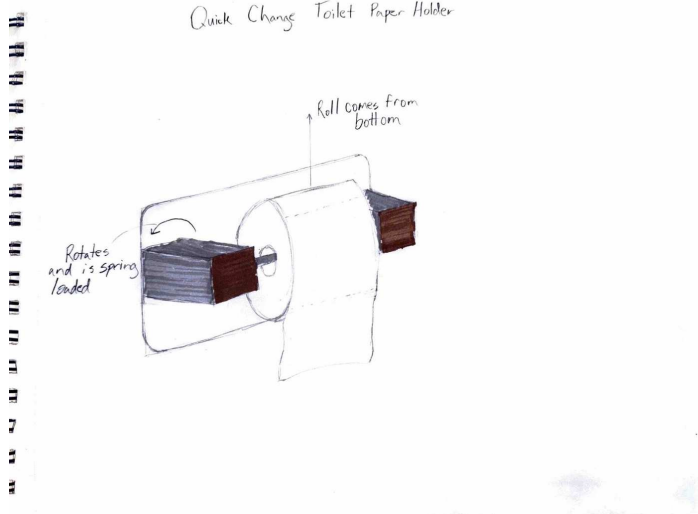
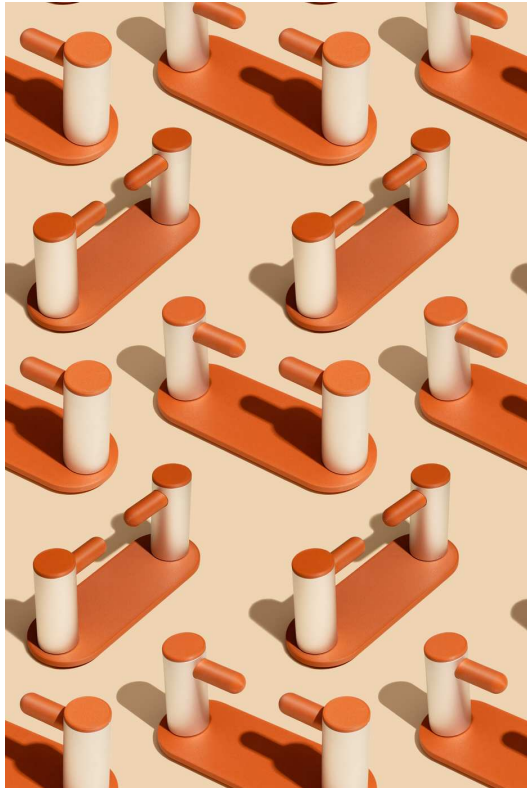
Two concurrent workstreams: prototype build support for the L35 small articulated loader and weld line optimization for the snowblower attachment line. Wrote initial standard work instructions for the L35, documenting part numbers, installation methods, and torque specifications. Designed a hydraulic hose routing jig for pre-assembly staging and created the first mating fixture to join the machine's two halves using PTC Creo. Participated in FMEA sessions. Conducted time studies on the snowblower line, identified cycle time imbalances, and redistributed weld tasks to level production flow. The L35 launched as Bobcat's most powerful small articulated loader.

Process Engineering, Fixture Design, PTC Creo, Lean, FMEA, Time Study

PROJECT 06

Roto Toilet Paper Holder

Product Design · April 2025

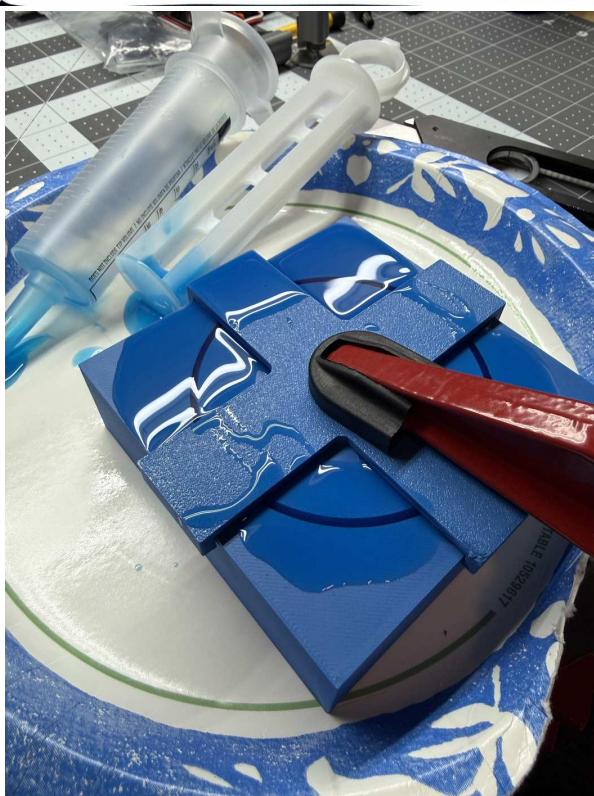
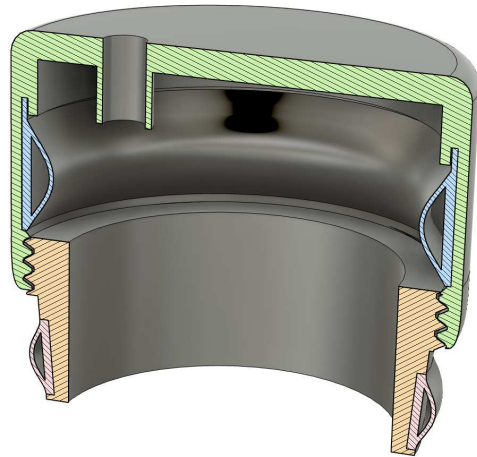
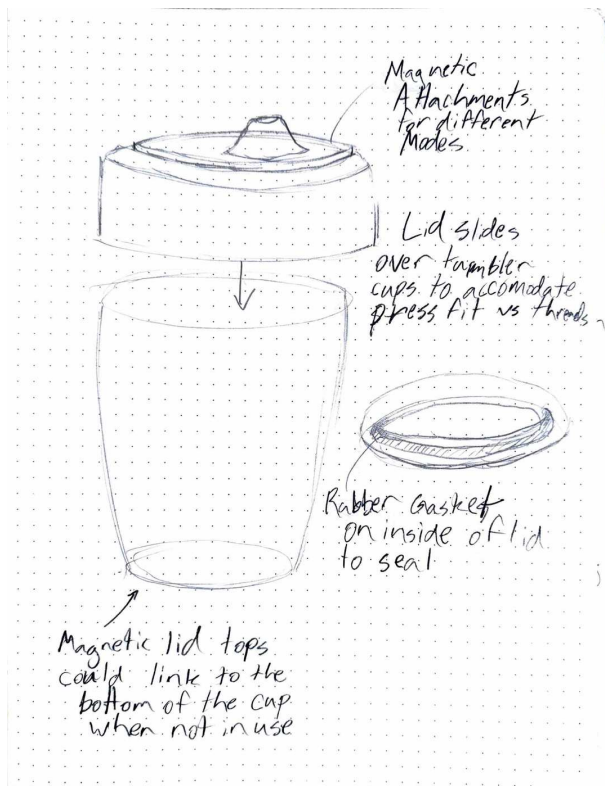


First consumer product and first product launch from Figura Studio, a home goods design company. Roto focuses on one-handed operation, making toilet paper roll changes faster, cleaner, and more accessible. Patent protection filed April 2025. Active ongoing program with future material iterations and collaborative versions planned with creators and brands.

Patent Pending, Consumer Product, DFM, Product Launch

Universal Lid

Product Design · February 2025



Universal lid designed to fit various tumblers and water bottles, solving the common problem of lost lids. Early prototypes used 3D-printed PLA with TPU gaskets squeezing onto bottle tops. Iterations progressed to overcast silicone for improved flexibility and seal. Developed through the Innovation Studio 1 program with rapid user testing cycles.

Prototyping, 3D Printing, Silicone Casting, User Testing